■ UZHAVU KAAPPAAN

Smart Crop Rotation & Farmer Soil Feeding Action Plan

Certified P025 Agronomic Model · 782,000+ Multi-Mandi Empirical Records

Farmer Name: Ramesh Kumar | **Contact:** +91 98421 54820
Farm Location: Coimbatore, Tamil Nadu | **Total Area:** 4.5 Acres
Soil Texture: Red Loam | **Irrigation:** Drip System (85% eff.)
Plan Ref ID: #UK-P025-FARM-101 | **Issue Date:** March 2026

1. Comprehensive Soil Health Diagnosis & Land Fertility Status

Overall Health Score	Land Fertility Classification	Primary Deficiencies & Required Action
63 / 100	**Moderate Nitrogen & Organic Carbon Depletion** Exhausted by 3 continuous tomato cycles	**Low Nitrogen:** 42 kg/ha (Target: 80-160 kg/ha) **Low Organic Carbon:** 0.52% (Target: >0.80%) **Zinc Deficient:** 0.6 ppm (Target: >1.0 ppm)

Nutrient Parameter	Measured Level	Optimal Range	Deficit / Status	Agronomic Impact & Crop Response
Nitrogen (N)	42.0 kg/ha	80 - 160 kg/ha	-38.0 kg/ha	Restricts leaf canopy; urgent pulse rotation needed
Phosphorus (P)	28.0 kg/ha	30 - 60 kg/ha	-2.0 kg/ha	Slightly sub-optimal root elongation
Potassium (K)	55.0 kg/ha	60 - 120 kg/ha	Sufficient	Adequate for disease resistance and pod fill
Soil pH	6.50	6.0 - 7.5	Optimal	Neutral; maximum nutrient bioavailability
Organic Carbon (OC)	0.52%	0.80 - 1.50%	-0.28% Deficit	Low moisture holding capacity & soil biology
Electrical Cond. (EC)	0.42 dS/m	< 1.0 dS/m	Normal	Non-saline root environment safe for legumes

2. Prescribed Soil Feeding & Fertilizer Dosage Schedule (for 4.5 Acres Land)

Feeding Stage	Recommended Inputs	Dose / Acre	Total (4.5 Ac)	Application Method & Agronomic Objective
Basal Land Prep (Prior to Sowing)	FYM / Vermicompost Neem Cake Bio-fertilizer (Rhizobium+PSB)	4.0 Tonnes 100 kg 2 kg + 2 kg	18.0 Tonnes 450 kg 9 kg + 9 kg	Broadcast & incorporate during final ploughing to boost Organic Carbon & inoculate root nodules
Sowing Time (Basal Starter)	DAP (Di-Ammonium Phos.) MOP (Potash) Zinc Sulphate ($ZnSO_4$)	25 kg 15 kg 10 kg	112.5 kg 67.5 kg 45.0 kg	Apply in bands 5cm below seed line for rapid root growth & zinc deficiency correction
Vegetative Growth (25-30 Days)	Urea (Top Dressing)	15 kg **(-30% saved)**	67.5 kg	Green gram naturally fixes 40 kg N/ha; reduced urea prevents lodging and saves input cost
Flowering & Pods (45-50 Days)	19:19:19 Soluble NPK Borax (0.2% Boron)	1.0 kg (Foliar) 200 g (Foliar)	4.5 kg 900 g	Foliar spray during early morning to prevent flower drop and promote uniform pod filling

3. Monoculture Penalty & Phytosanitary Risk Evaluation

■ CONTINUOUS TOMATO CULTIVATION DETECTED (3 Consecutive Seasons: 2023-2025)
- **Nutrient Depletion:** Prolonged solanaceous feeder demand without crop rotation depleted root-zone Nitrogen to 42 kg/ha.
- **Pathogen Risk:** Severe build-up of Early Blight (*Alternaria solani*) and Root-Knot Nematodes (*Meloidogyne spp.*).
- **Enforced Rotation:** Tomato is penalized by **-30%** in evaluation rankings to enforce immediate pulse restoration.

4. Recommended 3-Season Restorative Crop Rotation Plan

Season 1 (Immediate - Kharif)	Season 2 (Restoration - Rabi)	Season 3 (Stabilization - Zaid)
Green Gram (Moong) • **Crop Family:** Legume (N-Fixer) • **Duration:** 70-75 Days • **Seed Rate:** 8-10 kg / acre • **Mandi Price:** Rs. 85.00/kg (Live APMC) • **Est. Net Profit:** Rs. 33,500 / acre • **Biological Role:** Fixes 35-45 kg N/ha naturally and breaks tomato blight disease cycle.	**Groundnut (Arachis hypogaea)** • **Crop Family:** Legume (Restorer) • **Duration:** 105-110 Days • **Seed Rate:** 50-55 kg / acre • **Mandi Price:** Rs. 72.00/kg (Live APMC) • **Est. Net Profit:** Rs. 41,200 / acre • **Biological Role:** Deep rooting restores subsoil porosity and builds biomass organic carbon.	**Wheat / Maize (High Biomass)** • **Crop Family:** Poaceae (Cereal Feeder) • **Duration:** 115-120 Days • **Seed Rate:** 40 kg / acre • **Mandi Price:** Rs. 23.75/kg (Live APMC) • **Est. Net Profit:** Rs. 27,300 / acre • **Biological Role:** Stabilizes fixed nitrogen, provides rich fodder biomass, and completes recovery.

5. Multi-Season Financial Returns & Soil Health Recovery Trajectory

Rotation Stage	Crop Cultivated	Cost / Ac	Gross / Ac	Net Profit / Ac	Farm Total (4.5 Ac)	Soil Trajectory
Baseline	Depleted Tomato	Rs. 36,000	Rs. 48,000	Rs. 12,000	Rs. 54,000	**58 / 100** (Baseline)
Season 1 (Kharif)	Green Gram (Moong)	Rs. 12,500	Rs. 46,000	Rs. 33,500	Rs. 1,50,750	**65 / 100** (+7 recovery)
Season 2 (Rabi)	Groundnut	Rs. 18,000	Rs. 59,200	Rs. 41,200	Rs. 1,85,400	**72 / 100** (+7 recovery)
Season 3 (Zaid)	Wheat / Maize	Rs. 14,200	Rs. 41,500	Rs. 27,300	Rs. 1,22,850	**79 / 100** (Optimal)
3-Season Total	**Restorative Rotation**	**Rs. 44,700**	**Rs. 1,46,700**	**Rs. 1,02,000 / ac**	**Rs. 4,59,000 Total**	**79 / 100 (Restored)**

■ Extension Agronomist Mandatory Field Checklist:
1. **Bio-Inoculation:** Treat Green Gram seeds with *Rhizobium leguminosarum* bio-fertilizer @ 25g/kg seed before sowing.
2. **Input Cost Savings:** Do NOT exceed 15 kg Urea/acre; legume root nodules satisfy remainder requirements.
3. **Residue Retention:** Do not burn crop stubble. Incorporate haulms into the topsoil to rebuild Organic Carbon above 0.8%.
4. **Irrigation Scheduling:** Provide light irrigation at flowering and pod filling; avoid root waterlogging.

Certified by: UZHAVU KAAPPAAN P025 Model
Database: 737k Mandi Trades + 19k State Harvests + 2.2k Lab Sensor Records

Authorized Agronomist Extension Officer:

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Department of Agriculture & Extension, Coimbatore Division